

1.5 Solving Quadratic Equations - Part 2: Solve Using the Q.F. - Worksheet

MCR3U

SOLUTIONS

1) Use the discriminant to determine the number of roots for each quadratic equation.

a) $x^2 - 10x + 25 = 0$

$$b^2 - 4ac = (-10)^2 - 4(1)(25)$$

$$= 0$$

∴ 1 root

c) $2x^2 - 8x + 9 = 0$

$$b^2 - 4ac = (-8)^2 - 4(2)(9)$$

$$= -8$$

∴ No Roots

b) $3x^2 + 4x + \frac{4}{3} = 0$

$$b^2 - 4ac = (4)^2 - 4(3)\left(\frac{4}{3}\right)$$

$$= 0$$

∴ 1 root

d) $-2x^2 + 0.75x + 5 = 0$

$$b^2 - 4ac = (0.75)^2 - 4(-2)(5)$$

$$= 40.5625$$

∴ 2 Roots

2) Solve each quadratic using the quadratic formula. Give exact answers.

a) $x^2 + 4x - 3 = 0$

$$x = \frac{-4 \pm \sqrt{(4)^2 - 4(1)(-3)}}{2(1)}$$

$$x = \frac{-4 \pm \sqrt{28}}{2}$$

$$x = \frac{-4 \pm 2\sqrt{7}}{2}$$

$$x = \frac{2(-2 \pm \sqrt{7})}{2}$$

$$x_1 = -2 + \sqrt{7}$$

and

$$x_2 = -2 - \sqrt{7}$$

c) $x^2 = -5x + 2$

$$x^2 + 5x - 2 = 0$$

$$x = \frac{-5 \pm \sqrt{(5)^2 - 4(1)(-2)}}{2(1)}$$

$$x = \frac{-5 \pm \sqrt{33}}{2}$$

$$x_1 = \frac{-5 + \sqrt{33}}{2} \quad x_2 = \frac{-5 - \sqrt{33}}{2}$$

b) $-x^2 + 12 = 9x$

$$-x^2 - 9x + 12 = 0$$

$$-1(x^2 + 9x - 12) = 0$$

$$x^2 + 9x - 12 = 0$$

$$x = \frac{-9 \pm \sqrt{(9)^2 - 4(1)(-12)}}{2(1)}$$

$$x = \frac{-9 \pm \sqrt{129}}{2}$$

$$x_1 = \frac{-9 + \sqrt{129}}{2} \quad \text{or} \quad x_2 = \frac{-9 - \sqrt{129}}{2}$$

d) $x^2 - 3x + 1 = 6$

$$x^2 - 3x - 5 = 0$$

$$x = \frac{3 \pm \sqrt{(-3)^2 - 4(1)(-5)}}{2(1)}$$

$$x = \frac{3 \pm \sqrt{29}}{2}$$

$$x_1 = \frac{3 + \sqrt{29}}{2} \quad x_2 = \frac{3 - \sqrt{29}}{2}$$

$$e) x^2 + 6x + 9 = 0$$

$$x = \frac{-6 \pm \sqrt{(6)^2 - 4(1)(9)}}{2(1)}$$

$$x = \frac{-6 \pm \sqrt{0}}{2}$$

$$x = -3$$

$$g) 5x^2 - 3x - 1 = 0$$

$$x = \frac{3 \pm \sqrt{(-3)^2 - 4(5)(-1)}}{2(5)}$$

$$x = \frac{3 \pm \sqrt{29}}{10}$$

$$x_1 = \frac{3 + \sqrt{29}}{10} \quad x_2 = \frac{3 - \sqrt{29}}{10}$$

$$i) 4x^2 - 25 = 0$$

$$4x^2 + 0x - 25 = 0$$

$$x = \frac{0 \pm \sqrt{(0)^2 - 4(4)(-25)}}{2(4)}$$

$$x = \frac{\pm 20}{8}$$

$$x = \frac{5}{2} \quad x = -\frac{5}{2}$$

$$k) 8x^2 + 4x - 5 = 0$$

$$x = \frac{-4 \pm \sqrt{(4)^2 - 4(8)(-5)}}{2(8)}$$

$$x = \frac{-4 \pm \sqrt{176}}{16}$$

$$x = \frac{-4 \pm 4\sqrt{11}}{16}$$

$$x = \frac{4(-1 \pm \sqrt{11})}{16}$$

$$x_1 = \frac{-1 + \sqrt{11}}{4} \quad x_2 = \frac{-1 - \sqrt{11}}{4}$$

$$f) 4x^2 - 6x - 1 = 0$$

$$x = \frac{6 \pm \sqrt{(-6)^2 - 4(4)(-1)}}{2(4)}$$

$$x = \frac{6 \pm \sqrt{52}}{8}$$

$$x = \frac{6 \pm 2\sqrt{13}}{8}$$

$$x = \frac{3 \pm \sqrt{13}}{4}$$

$$x = \frac{3 \pm \sqrt{13}}{4}$$

$$x_1 = \frac{3 + \sqrt{13}}{4} \quad x_2 = \frac{3 - \sqrt{13}}{4}$$

$$h) -x^2 + 7x - 18 = 0$$

$$x = \frac{-7 \pm \sqrt{(7)^2 - 4(-1)(-18)}}{2(-1)}$$

$$x = \frac{-7 \pm \sqrt{-23}}{-2}$$

$$\therefore \text{No Roots}$$

$$j) 3x^2 - 7x - 4 = x^2 - 4x$$

$$2x^2 - 3x - 4 = 0$$

$$x = \frac{3 \pm \sqrt{(-3)^2 - 4(2)(-4)}}{2(2)}$$

$$x = \frac{3 \pm \sqrt{41}}{4}$$

$$x_1 = \frac{3 + \sqrt{41}}{4} \quad x_2 = \frac{3 - \sqrt{41}}{4}$$

$$l) 4x^2 - 18x = 0$$

$$x = \frac{18 \pm \sqrt{(-18)^2 - 4(4)(0)}}{2(4)}$$

$$x = \frac{18 \pm 18}{8}$$

$$x = \frac{18+18}{8} \quad x = \frac{18-18}{8}$$

$$x_1 = \frac{9}{2} \quad x_2 = 0$$

3) Solve each quadratic equation using any method

a) $3x^2 - 12x = 0$

$$3x(x-4) = 0$$

$$3x = 0$$

$$x = 0$$

$$x-4 = 0$$

$$x_2 = 4$$

b) $2x^2 + 4x - 6 = 0$

$$S = 4$$

$$P = -12$$

$$6 \text{ and } -2$$

$$2x^2 + 6x - 2x - 6 = 0$$

$$(2x^2 + 6x) + (-2x - 6) = 0$$

$$2x(x+3) - 2(x+3) = 0$$

$$(x+3)(2x-2) = 0$$

$$x_1 = -3$$

$$x_2 = 1$$

c) $3x^2 + 5x - 2 = 0$

$$S = 5$$

$$P = -6$$

$$6 \text{ and } -1$$

$$3x^2 - 1x + 6x - 2 = 0$$

$$x(3x-1) + 2(3x-1) = 0$$

$$(3x-1)(x+2) = 0$$

$$x_1 = \frac{1}{3} \quad x_2 = -2$$

d) $4x^2 - 11x - 8 = 0$

$$x = \frac{11 \pm \sqrt{(-11)^2 - 4(4)(-8)}}{2(4)}$$

$$x = \frac{11 \pm \sqrt{249}}{8}$$

$$x_1 = \frac{11 + \sqrt{249}}{8} \quad x_2 = \frac{11 - \sqrt{249}}{8}$$

4) Three lengths of pipe measuring 24 cm, 31 cm, and 38 cm will be used to create a right triangle. The same length of pipe will be cut off each of the three pipes to allow a right triangle to be created. What is that length?

$$(24-x)^2 + (31-x)^2 = (38-x)^2$$

$$576 - 48x + x^2 + 961 - 62x + x^2 = 1444 - 76x + x^2$$

$$x^2 - 34x + 93 = 0$$

$$(x-3)(x-31) = 0$$

$$x = 3$$

$$x = 31$$

↑
reject because can't
cut 31 cm from 24 cm.

$$3 \text{ cm}$$

Answers

1) a) one b) one c) none d) two

2) a) $-2 + \sqrt{7}, -2 - \sqrt{7}$ b) $\frac{-9-\sqrt{129}}{2}, \frac{-9+\sqrt{129}}{2}$ c) $\frac{-5+\sqrt{33}}{2}, \frac{-5-\sqrt{33}}{2}$ d) $\frac{3+\sqrt{29}}{2}, \frac{3-\sqrt{29}}{2}$ e) -3 f) $\frac{3+\sqrt{13}}{4}, \frac{3-\sqrt{13}}{4}$

g) $\frac{3+\sqrt{29}}{10}, \frac{3-\sqrt{29}}{10}$ h) no roots i) $\frac{5}{2}, -\frac{5}{2}$ j) $\frac{3+\sqrt{41}}{4}, \frac{3-\sqrt{41}}{4}$ k) $\frac{-1+\sqrt{11}}{4}, \frac{-1-\sqrt{11}}{4}$ l) $\frac{9}{2}, 0$

3) a) $x = 0$ and $x = 4$ b) $x = 1$ and $x = -3$ c) $x = \frac{1}{3}$ and $x = -2$ d) $x = \frac{11 \pm \sqrt{249}}{8}$

4) 3 cm